Index-Translate: A Multilingual Translation Model Family

Text, Speech, Controlled Dubbing, and Long-Document Translation

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Abstract

We introduce Index-Translate, a multilingual translation model family that combines a shared multilingual foundation with specialized training for general translation, instruction following, speech translation, controlled dubbing, and long-document translation. It includes three model sizes, 2B, 9B, and 35B-A3B, and supports translation in 150 languages, with multilingual instruction following. Evaluations on general translation and complex translation instructions show that Index-Translate outperforms translation models of comparable size and achieves performance comparable to 100B-scale translation models and frontier models. Image-text evaluation shows that translation remains possible without image examples in Index-Translate-specific training, although results vary by model size. Index-Translate-Omni supports text and image translation, Chinese and English speech recognition, and bidirectional Chinese–English speech translation in a single checkpoint at each scale. Index-Echo provides end-to-end speech-to-text and speech-to-speech translation, outperforming existing end-to-end models and achieving performance comparable to frontier omni models. Index-Homura extends the family to syllable-controlled dubbing. Index-NativeLong introduces native long-document translation with a dedicated task formulation and benchmark. These capabilities support diverse translation tasks, including multilingual content production.

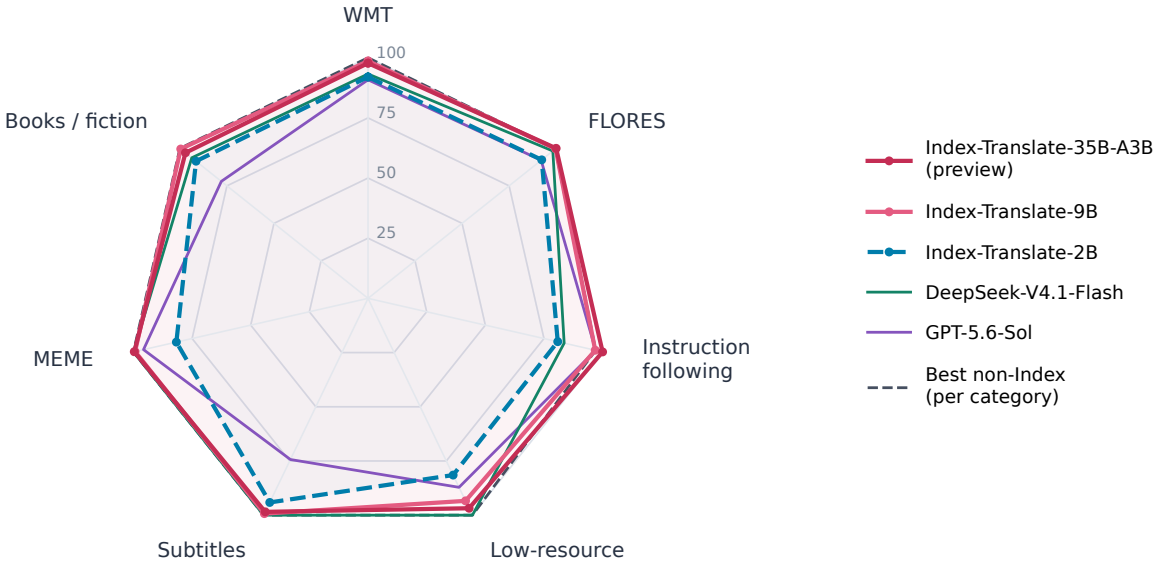


Figure 1: Seven-category text-translation comparison using the online demo's updated aggregates. Each axis is min–max normalized to 0–100 over all 14 models, including 35B-A3B (preview). Instruction following averages instTrans and IFMT-Bench IFscore; subtitles average OpenSubtitles and MuST-Cinema; books average OPUS-Books and GuoFeng-Webnovel. The gray dashed curve combines the best non-Index score per category.

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1 Introduction

Multilingual translation models must preserve meaning across languages while accommodating differences in task requirements, input and output modalities, and context length. Index-Translate addresses these needs as a model family built on a shared multilingual foundation.

Content production provides concrete examples of these requirements. Structured documents must preserve fields and terminology; community expressions require cultural interpretation; dubbed utterances must fit a target length; and books must maintain identities across distant passages. Speech translation adds acoustic understanding and spoken output.

Our central question is which of these capabilities compact translation models can share and consolidate, and which benefit from dedicated training or output interfaces. Starting from Qwen3.5 base models before post-training [1], Index-Translate addresses this question through three connected design choices:

- **A shared multilingual foundation.** We combine general, monolingual, parallel, and pivot-organized text to build a translation foundation for 150 languages, studying how language coverage and data organization affect translation quality and general capability retention.
- **Integration of complementary translation skills.** General, instruction-following, and meme experts use task-specific SFT and RL, with adversarial supervision for judge-based rewards. Parameter interpolation combines these experts, followed by targeted multi-teacher on-policy distillation (MOPD) for tasks that remain weak after merging.
- **Dedicated training for speech, length control, and long documents.** Index-Echo connects the text foundation to speech perception and generation; Index-Homura learns explicit syllable control; and Index-NativeLong addresses native long-document translation with a dedicated task formulation and benchmark. These adaptations address different input, output, and sequence-length requirements.

Our analyses examine both gains and limits of capability sharing. Broader language coverage improves non-core translation, while expert integration exposes trade-offs among quality, instruction adherence, and cultural adaptation. Precise syllable control transfers poorly through the studied merging and distillation settings, motivating a dedicated specialist. Text-only adaptation also supports image-text translation through the inherited visual pathway, with stronger cross-modal generalization at larger scales (Section 4.3). Table 1 summarizes the model family; Appendix C lists the supported languages and their identifiers.

Model	Backbone	Task
Index-Translate	2B / 9B / 35B-A3B	Multilingual text and image-text translation, with translation instructions
Index-Translate-Omni	2B / 9B / 35B-A3B	Text and image translation, Chinese and English speech recognition, and Chinese–English speech-to-text translation
Index-Echo S2TT	2B / 9B	Speech-to-text translation
Index-Echo S2ST	2B / 9B	Speech-to-speech translation with voice conditioning
Index-Homura	2B / 9B	Translation with a specified syllable count
Index-NativeLong	2B / 9B	Native long-document translation

Table 1: Index-Translate model family and evaluated model sizes.

2 Training

2.1 Multilingual mid-training

Data design. Mid-training improves cross-language alignment while retaining broad linguistic competence. We combine general pre-training replay, independently sampled monolingual text, and parallel translations.

Parallel and pivot organization. Aligning the same content across languages can improve multilingual consistency and performance [2]. We distinguish ordinary parallel text (P) from pivot-organized text (V). In the

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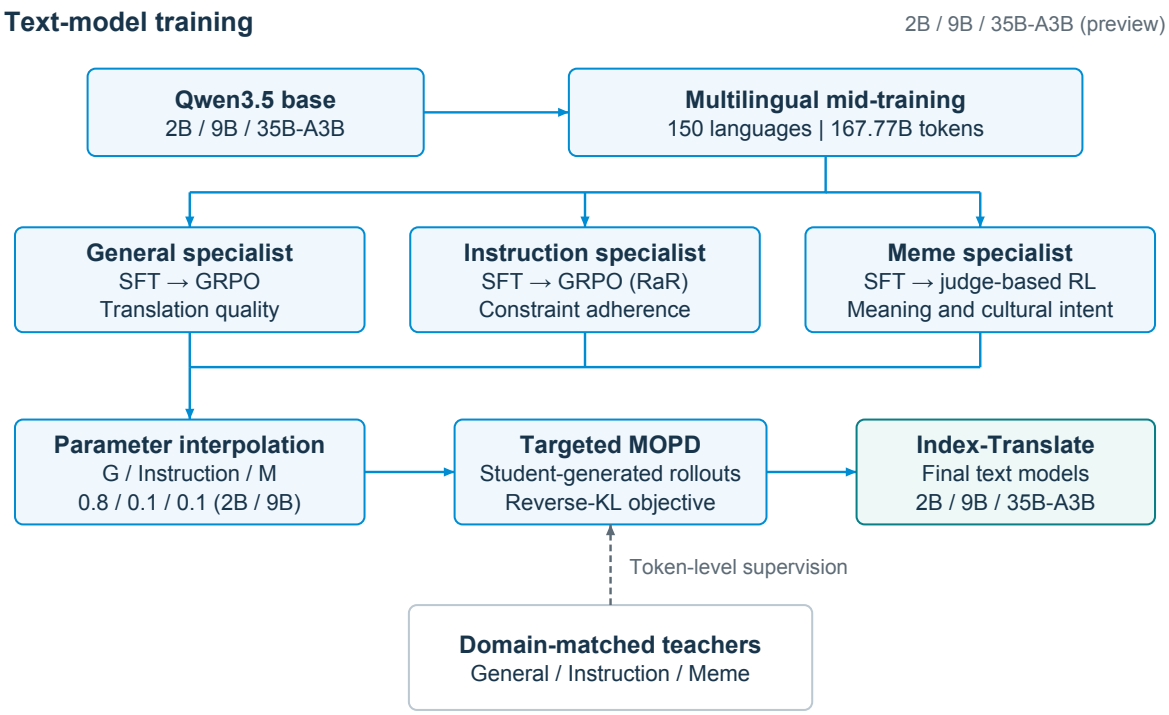


Figure 2: Training overview of the text-model family, including 2B, 9B, and 35B-A3B. General, Instruction, and Meme specialists share a multilingual mid-trained initialization. Parameter interpolation is followed by targeted MOPD, using domain-matched teachers to supervise student-generated rollouts with a reverse-KL objective, to obtain the final models. The displayed interpolation weights apply to the evaluated 2B and 9B models. Judge-based rewards use RIVAL supervision as described in Section 2.2.

Qwen3-8B pilot, ordinary parallel examples align two languages directly, while pivot groups connect multiple language variants through an English anchor. Parallel degree counts the languages aligned to the same content; data-stream weights determine how often each construction is sampled.

The final recipe uses separate pivot streams for core and full language coverage. The stream labels `pivot-20` and `pivot-148` in Table 2 exclude English and Chinese from their counts; including these two languages gives 22 core and 150 total entries in Appendix C. Ordinary parallel text enters the constant stage with general and monolingual text; pivot data dominate decay. Appendix A.1 illustrates the two organizations.

Training recipe. The shared 2B, 9B, and 35B-A3B recipe follows the stable and decay phases of the warmup–stable–decay schedule [3]. We use AdamW, a sequence length of 4,096, a global batch of 4,096 sequences, and a micro-batch of 128. Table 2 gives the stage mixtures, learning rates, and token budgets. The long-document extension increases sequence length during decay (Section 3.3).

Stage	Sampling ratio	Steps	Tokens (B)	Learning rate
Constant	general : parallel : monolingual $= 1{:}1{:}1$	7,000	117.44	5×10^{-5}
Decay	general : pivot-20 : pivot-148 $= 1{:}4{:}2$	3,000	50.33	$5 \times 10^{-5} \rightarrow 5 \times 10^{-6}$
Total		10,000	167.77	

Table 2: Base translation mid-training recipe shared by 2B, 9B, and 35B-A3B. Token budgets use 4,096 tokens per sequence and 4,096 sequences per global batch.

Section 5.1 analyzes recipe choices, stage progression, and language coverage.

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2.2 Post-training

We develop general translation, instruction-following, and meme-translation experts through a shared SFT-to-RL workflow. Each expert uses task-specific data and supervision; Section 2.3 describes how we combine their complementary capabilities.

Supervised fine-tuning. General translation SFT uses multilingual training data with translations regenerated by frontier models and filtered for language consistency and quality, retaining both core and low-resource directions. Instruction SFT uses 1.045 million examples across ten real-world scenarios, with constraints on structure, terminology, context, and style. It mixes instruction translation and general-capability data at a $1{:}1$ token ratio. Meme SFT adds curated examples that teach the model to resolve nonliteral meanings and preserve pragmatic intent through idiomatic paraphrase, explanation, or cultural adaptation.

Reinforcement learning. RL refines the SFT models with rewards tailored to each task. General and instruction translation use GRPO. General translation combines reference-based XCOMET-XXL with target-language validity and source-grounded adequacy judgments from Gemini 2.5 Flash. Instruction translation uses Rubric-as-Reward (RaR), with Gemini 3.1 Pro judging instruction compliance. Using g for binary gates and q for graded scores, the rewards are

$$R_{\text{gen}} = g_{\text{lang}}\, q_{\text{XCOMET}}\, q_{\text{adeq}}, \tag{1}$$

$$R_{\text{inst}} = g_{\text{lang}}\, g_{\text{hard}}\, \big(q_{\text{qual}} + q_{\text{soft}}\big). \tag{2}$$

Here g_{lang} denotes the task-specific target-language gate, and g_{hard} checks required structure, terminology, and preservation of code or placeholders; a failed check sets the corresponding gate to zero. For general translation, q_{XCOMET} is the reference-based XCOMET-XXL score and q_{adeq} is the adequacy score applied as a multiplicative discount for omissions, additions, and other fidelity errors. For instruction translation, q_{qual} is the rubric-based translation-quality score and q_{soft} averages soft-constraint scores for style and contextual interpretation. Timed-subtitle constraints also assess agreement between syllable counts and segment durations.

Meme RL uses a reference-guided LLM judge to assess contextual meaning, cultural function, and pragmatic effect.

Adversarial judge supervision. We supplement judge-based rewards with our RIVAL framework [4]. Training alternates reward-model updates on current policy outputs with policy optimization against the refreshed reward. Preference ranking, auxiliary quantitative supervision, and replay of earlier outputs help the evaluator track evolving failure patterns. In our earlier Qwen2.5-7B study [4], subtitle quality judged by GPT-4o rose from 3.26 to 3.68 between RIVAL iterations 0 and 1; training-curve analysis also showed reduced divergence between reward-model and external evaluations. At iteration 2 on WMT English-to-Chinese, adding quantitative supervision to preference ranking improved BLEU from 30.14 to 39.39 and COMETKiwi from 71.91 to 72.60. These results motivate using adaptive reward supervision to mitigate reward hacking. Appendix B.5 gives representative cases from Index-Translate training.

2.3 Expert integration

Parameter interpolation. We first combine compatible specialists in parameter space. For the general (G), instruction, and meme (M) specialists, we use

$$\theta_{\text{merge}} = w_G\theta_G + w_{\text{Instruction}}\theta_{\text{Instruction}} + w_M\theta_M, \qquad w_G + w_{\text{Instruction}} + w_M = 1. \tag{3}$$

Parameter averaging follows the model-soup approach [5] and produces a merged initialization for targeted MOPD. The interpolation stage for the evaluated Index-Translate-2B and 9B models uses $w_G/w_{\text{Instruction}}/w_M = 0.8/0.1/0.1$. We select weights using both translation quality and instruction following (Section 5.2), then use post-merge evaluation to identify task types for targeted MOPD.

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Multi-teacher on-policy distillation. Following the MOPD framework [6], we use domain-matched teachers to supervise student-generated rollouts with a reverse-KL objective. In our workflow, this serves as a targeted follow-up to parameter interpolation, addressing task types where the merged model shows weaknesses. Targeted MOPD is part of the training pipeline for the final Index-Translate models. Section 5.2 reports configuration studies using an earlier 2B reference merge, with general, instruction, and meme sampling ratios of $1{:}1{:}1$ or $1{:}2{:}1$, including a configuration using three 9B specialists as teachers.

3 Model Extensions

3.1 Index-Echo: speech translation

Speech-to-text translation. Index-Echo connects a Qwen3-Omni AuT audio encoder [7] and connector to an Index-Translate 2B or 9B decoder. We train the resulting model end to end on speech translation, combining acoustic input, target-language instructions, and translated text in one sequence. This design uses the multilingual text model as the translation component while learning to interpret the source directly from audio.

Speech-to-speech translation. We initialize S2ST training from the trained 2B and 9B S2TT models. The speech-generation path replaces the CosyVoice3 [8] text-tokenizer interface with a roughly 30M-parameter Hidden2CV mapper from the translator's final hidden states to the semantic layer. We first distill the mapper with the S2TT backbone and speech generator frozen. The S2TT backbone remains frozen during subsequent differentiable reward optimization (DiffRO). Following CosyVoice3 [8], DiffRO backpropagates a Token2Text content-consistency reward through Gumbel-Softmax speech-token samples. Source audio supplies a reference prompt and CampPlus speaker embedding for voice conditioning.

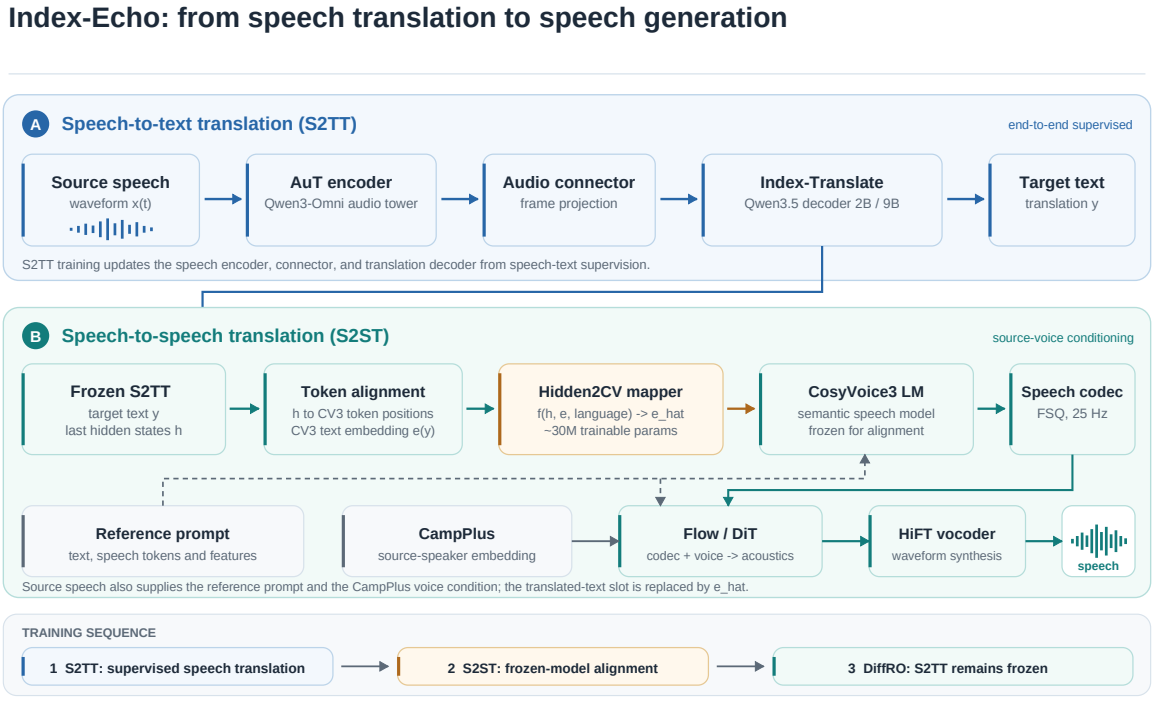


Figure 3: Index-Echo speech-to-text and speech-to-speech architecture. S2ST training aligns the mapper and then applies DiffRO; the S2TT backbone remains frozen throughout.

The resulting S2ST models cover six translation directions: Chinese to English, Spanish, and Japanese, and English to Chinese, Spanish, and Japanese. The target language is specified explicitly, and source audio provides voice conditioning.

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3.2 Index-Homura: translation for dubbing

Index-Homura adapts the 2B and 9B text models to translation with an explicitly specified target syllable count. Given a source text and a target count $s^\star$, the model adjusts its wording to meet the requested count while preserving meaning and natural expression.

Building on the reinforcement-learning approach of HOMURA [9], we use GRPO to jointly optimize translation quality and adherence to the specified syllable count. The length reward measures how closely the generated translation matches $s^\star$, enabling direct control over the syllable count of the output.

3.3 Index-NativeLong: native long-document translation

Book translation requires faithful coverage and continuity across distant passages. We study *native long-document translation*, in which a model receives the full source document and produces its complete translation in a single generation. Index-NativeLong extends our multilingual text foundation to this task, combining long-context understanding with sustained output [12].

Why translate the full document? Glossary-assisted frameworks such as TranslateBooksWithLLMs [30] connect chunks through terminology lists and neighboring context, but can still lose conceptual distinctions and document-wide consistency. In our glacial-history case, 9B preserves the distinction between an *ice age* (冰河时代) and its *glacials* (冰期), while chunked Qwen3.8-Flash conflates them even with a full-source glossary. For recurring names such as 王妃 and Zeitlin, native 9B maintains consistent translations where the same model with glossary-assisted chunking changes them. Appendix B.4 gives the original excerpts and comparison settings.

Native generation keeps the full source and translation history together, supporting these document-level requirements without separate glossary preparation or repeated context input. We introduce NativeLongBench to evaluate this capability; Section 4.7 presents the data and results.

Training. We adapt the 2B and 9B models through full-parameter SFT on aligned book passages and complete target translations. The 2B model follows 128K-sequence mid-training decay and general SFT; the 9B model continues from the integrated model in Section 2.3. Both receive bidirectional document supervision over approximately 4K–64K Chinese-side tokens. GuoFeng ablations support long-sequence decay and show comparable overall quality for RoPE and NoPE after long-document SFT (Appendix A.3).

3.4 Index-Translate-Omni: text, image, and speech

Index-Translate-Omni packages text and image translation, automatic speech recognition (ASR) in Chinese and English, and bidirectional Chinese–English speech-to-text translation in one checkpoint at each of the 2B, 9B, and 35B-A3B scales. The translation backbone produces text for each input modality. Index-Echo S2ST is the family's speech-to-speech model.

The released 35B-A3B model uses one shared audio encoder and connector across all four speech tasks. We interpolate the parameters of both components using $\theta_{\text{audio}} = 0.8\theta_{\text{native}} + 0.2\theta_{\text{zh}}$, where θ_{zh} is the front end adapted for Chinese ASR and θ_{native} is the original audio front end. The language and vision weights stay fixed during this interpolation. The native checkpoints and usage instructions are available in the Hugging Face and ModelScope collections.

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4 Evaluation

4.1 Evaluation setup

Unless a training stage is explicitly identified, all Index-Translate-2B and 9B evaluations use the same final released models, obtained by parameter interpolation with general, instruction, and meme weights of $0.8{:}0.1{:}0.1$, followed by targeted MOPD. We additionally evaluate Index-Translate-35B-A3B (preview). All evaluated models in the text translation and general-capability comparisons are run in non-thinking mode, with thinking explicitly disabled when supported. Speech baselines evaluated with thinking enabled are explicitly labeled in Table 8. Text translation uses greedy decoding and a 16,384-token context window, with the WMT26 settings specified below. Training-stage studies and model extensions report their settings separately. Table 3 summarizes the translation benchmarks, including FLORES-200 [13], WMT24++ [14], IFMTBench [15], and GuoFeng-Webnovel [16].

Benchmark	Items	Coverage and split
FLORES-200	126,000	420 directions among 21 languages; devtest; 300 per direction
FLORES low-resource pairs	104,000	1,040 directions among 62 languages; devtest; 100 per direction
WMT24++	6,000	English to 20 languages; test; 300 per direction
WMT26	4,897	23 directions covering 20 target languages
instTrans	3,000	Chinese to 20 languages; aligned evaluation set
instTrans low-resource	2,793	Chinese / English to low-resource languages; aligned evaluation set
IFMTBench	7,064	Translation instructions; preprocessing in Appendix B.6
OPUS-Books	2,700	English to nine languages; train split; 300 per direction
WMT Biomedical	600	English paired bidirectionally with six languages; test; 50 per direction
MTNT	4,013	English paired bidirectionally with French and Japanese; test
MuST-Cinema	3,803	English to seven languages; Amara test split
GuoFeng-Webnovel	25 chapters	Chinese to English; document-level evaluation
MEME	3,638	Culturally marked translation examples

Table 3: Text translation evaluation sets. Counts refer to evaluated inputs; GuoFeng counts chapters.

Instruction evaluation uses GPT-5.6-Sol and reports translation quality and instruction following separately. IFMTBench reports instruction compliance and reference-based XCOMET-XXL; its preprocessing and scoring details are given in Appendix B.6.

Reference-based translation metrics are COMET-22 (`wmt22-comet-da`) [17] and XCOMET-XXL [20]. FLORES scores average directions equally; Vertical averages the five domain benchmarks in Table 5. WMT26 uses reference-free GPT-5.6-Sol judgments on a 0–100 scale, averaged over 20 target languages, with greedy decoding, zero-shot Chinese instructions, a 3,584-token context, and a 1,536-token output limit.

4.2 Text translation

Table 4 summarizes text translation performance. Panel (b) includes low-resource general translation (FLORES_minor_pair) and instruction following (instTrans_minor), with off-target rates reported for both. Index-Translate-35B-A3B (preview) achieves the highest FLORES COMET-22 and instTrans IFscore among the compared models (0.8794 and 0.8336), with an instTrans quality score of 0.6901 and a WMT26 judge score of 76.76. Index-Translate-9B remains competitive on general translation and instruction following. The 2B model outperforms Hy-MT2-1.8B on all metrics available for both models.

Low-resource translation and instruction following. FLORES_minor_pair covers 104,000 inputs across 1,040 directions among 62 languages; instTrans_minor adds 2,793 instruction-translation tasks from Chinese or English to low-resource languages. Table 4(b) reports translation quality, instruction following, and off-target rates (the percentage of outputs not in the target language; lower is better). Among the three Index-Translate models, 35B-A3B (preview) achieves the highest low-resource FLORES COMET-22 and XCOMET-XXL scores (0.8168 and 0.7164), with a 2.4% off-target rate. On instTrans_minor, Index-Translate-9B achieves the highest IFscore (0.7725) and lowest off-target rate (3.47%) among all compared models; its quality score is 0.5222.

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(a) Main translation benchmarks

Model	FLORES	WMT24++	WMT26	instTrans		IFMTBench		Vertical	MEME
	COMET-22	COMET-22	Judge	Quality	IFscore	XCOMET-XXL	IFscore	mean	
Index-Translate-35B-A3B (preview)	**0.8794**	0.8586	76.76	0.6901	**0.8336**	0.7926	0.8991	0.8438	0.7405
Index-Translate-9B	0.8789	0.8601	75.35	0.6771	0.8209	0.7957	0.8760	0.8451	0.7387
Index-Translate-2B	0.8655	0.8489	60.26	0.5391	0.7569	0.7712	0.7584	0.8377	0.6443
Hy-MT2-1.8B	0.8522	0.8401	49.35	0.3181	0.4932	0.7493	0.7161	0.8314	0.3643
Hy-MT2-7B	0.8747	0.8593	60.51	0.5143	0.6079	0.8049	0.8741	0.8335	0.5139
Hy-MT2-30B-A3B	0.8787	**0.8624**	66.81	0.5725	0.6415	**0.8177**	0.9029	**0.8459**	0.5812
TranslateGemma-12B	0.8732	0.8524	71.19	0.4515	0.3068	0.8023	0.2892	0.8347	0.4281
North-Small-Translate	0.8784	0.8578	68.37	0.5697	0.5294	0.7657	0.8635	0.8357	0.6836
Qwen3.5-2B	0.6983	0.6933	32.11	0.0999	0.2431	0.6197	0.3836	0.7557	0.2062
Qwen3.5-9B	0.8316	0.8073	60.31	0.2467	0.0609	0.7341	0.5980	0.8199	0.5728
Qwen3.5-35B-A3B	0.8570	0.8290	71.33	0.3690	0.5204	0.7589	0.7822	0.8267	0.6447
DeepSeek-V4.1-Flash	0.8762	0.8510	83.55	0.6068	0.6374	0.7817	0.9090	0.8432	**0.7424**
GPT-5.6-Sol	0.8650	0.8469	**89.10**	**0.6902**	0.7624	0.7946	**0.9367**	0.8311	0.7194
Gemini 3.5 Flash Lite	0.8750	0.8497	79.52	0.6068	0.6374	0.7764	0.8854	0.8131	0.7034

(b) Low-resource translation and instruction following

Model	FLORES_minor_pair			instTrans_minor		
	COMET-22 ↑	XCOMET-XXL ↑	off-target ↓	Quality ↑	IFscore ↑	off-target ↓
Index-Translate-35B-A3B (preview)	0.8168	0.7164	2.4%	0.5151	0.7715	4.05%
Index-Translate-9B	0.7992	0.6805	4.0%	0.5222	**0.7725**	**3.47%**
Index-Translate-2B	0.7377	0.4817	4.2%	0.3050	0.6586	3.97%
Hy-MT2-1.8B	0.3163	0.2014	54.2%	0.0412	0.1313	65.27%
Hy-MT2-7B	0.4626	0.3334	35.7%	0.1121	0.2405	45.40%
Hy-MT2-30B-A3B	0.6746	0.5359	14.5%	0.2246	0.4449	15.47%
TranslateGemma-12B	0.8021	0.6217	1.7%	0.2412	0.2682	5.37%
North-Small-Translate	0.6855	0.4919	13.2%	0.2436	0.2902	26.89%
Qwen3.5-2B	0.3716	0.1852	39.1%	0.0069	0.0549	73.54%
Qwen3.5-9B	0.6774	0.4502	11.2%	0.1318	0.3532	18.51%
Qwen3.5-35B-A3B	0.7854	0.6257	3.5%	0.2874	0.4458	11.60%
DeepSeek-V4.1-Flash	**0.8333**	**0.7297**	**1.3%**	0.4793	0.5854	5.73%
GPT-5.6-Sol	0.7669	0.6918	10.4%	**0.5757**	0.6866	7.73%
Gemini 3.5 Flash Lite	0.8122	0.6995	3.3%	0.3927	0.5584	12.18%

Table 4: Text translation results for Index-Translate and comparison systems. Instruction judging uses GPT-5.6-Sol; quality and IFscore are separate metrics. WMT26 judge scores use a 0–100 scale, off-target rates are percentages, and other scores use 0–1. Vertical is the equal-weight mean of five domain scores. North-Small-Translate denotes the 218B-A25B model, and TranslateGemma denotes `translategemma-12b-it`. Panel (b) reports low-resource general translation and instruction following. Bold marks the best value in each column: lower is better for off-target rates, and higher is better for all other metrics.

Vertical-domain translation. Table 5 expands the five-domain Vertical mean. Index-Translate-9B leads the compared models on MuST-Cinema.

instTrans benchmark. We introduce instTrans to evaluate translation under task-specific instructions across subtitles, social content, articles, academic documents, books, and fiction. The evaluated core set contains 3,000 examples from Chinese to 20 target languages; a low-resource extension adds 2,793 examples from Chinese or English to low-resource languages. The benchmark covers ten constraint types, comprising five hard checks and five graded soft constraints. These include format and terminology preservation, subtitle syllable alignment, style consistency, cross-sentence terminology, coreference resolution, and context disambiguation. Multiple constraints can apply to the same example.

We report translation quality and IFscore separately. Quality uses three GPT-5.6-Sol judge calls per example on a 0/0.5/1 scale. IFscore gates the mean soft-constraint score by hard-constraint checks, assigning zero if any required hard check fails, then averages over examples. Valid quality judgments cover 2,961 / 2,960 core examples and 2,777 / 2,776 low-resource examples for 2B / 9B; means use valid judgments. Table 24 breaks down selected constraints, and Appendix B provides representative translation cases.

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Model	OPUS-Books	Biomedical	MTNT	MuST-Cinema	GuoFeng
Index-Translate-35B-A3B (preview)	0.8013	0.8681	0.8244	0.8962	0.8291
Index-Translate-9B	0.8010	0.8674	0.8246	**0.8979**	**0.8345**
Index-Translate-2B	0.7845	0.8635	0.8132	0.8930	**0.8345**
Hy-MT2-1.8B	0.7773	0.8592	0.8080	0.8838	0.8286
Hy-MT2-7B	0.7954	0.8671	0.8245	0.8929	0.7876
Hy-MT2-30B-A3B	**0.8054**	**0.8686**	0.8279	0.8966	0.8310
TranslateGemma-12B	0.8006	0.8673	0.8107	0.8886	0.8062
North-Small-Translate	0.8051	0.8667	0.8223	0.8936	0.7907
Qwen3.5-2B	0.6375	0.8296	0.7203	0.7951	0.7959
Qwen3.5-9B	0.7485	0.8618	0.7917	0.8728	0.8247
Qwen3.5-35B-A3B	0.7745	0.8645	0.8063	0.8839	0.8044
DeepSeek-V4.1-Flash	0.7962	0.8674	**0.8290**	0.8960	0.8274
GPT-5.6-Sol	0.7790	0.8676	0.8255	0.8705	0.8127
Gemini 3.5 Flash Lite	0.7845	0.8675	0.8238	0.8922	0.6977

Table 5: Vertical-domain translation results (higher is better): COMET-22 for the four sentence-level benchmarks and document-COMET for GuoFeng-Webnovel. Biomedical denotes the WMT Biomedical Translation Task. The equal-weight mean of these five columns is the Vertical score in Table 4. Bold marks the best score in each column.

Meme benchmark. MEME evaluates Chinese-to-English translation of culturally marked user-generated content, with 3,638 cases covering 703 terms and 857 disambiguated senses. Community experts validate sense assignments, and bilingual experts annotate translation strategies and quality-tiered references. Gemini-2.5-Flash receives each term's definition, usage examples, and references, assigning 1 for an accurate and idiomatic translation, 0.5 for a flawed but usable translation, and 0 for an unusable translation. Appendix B.2 illustrates how the models recover playful spellings, community aliases, and laughter expressions.

4.3 Image-text translation

Index-Translate inherits its visual pathway from the Qwen3.5 base models before post-training [1]; its own mid- and post-training use no images. We evaluate image-text translation on three probes: the public MMTIT-Bench [18] release (1,243 images in 14 source languages, translated into English and Chinese), 1,200 plain-background images rendered from FLORES_minor_pair text (60 source languages, two targets), and Vistra [19] (763 natural images, four targets). For Vistra, we join segment references in reading order to score whole-image output; this is an adapted protocol rather than an official benchmark score.

Each model receives only the image and a zero-shot translation instruction; source transcriptions and references are withheld from inference and used only for scoring. We compare Index-Translate 2B, 9B, and 35B-A3B (preview) with corresponding post-trained Qwen3.5 models. All use non-thinking decoding with temperature zero, top-p one, a 24,576-token context window, and a 4,096-token output limit. We report reference-based COMET-22 [17]; the metric evaluates translated text rather than visual layout or text replacement in the image.

Probe	Size	Index	Qwen3.5	Δ	Paired 95% CI
MMTIT	2B	0.5204	0.6826	-0.1622	$[-0.1705, -0.1538]$
MMTIT	9B	0.7347	0.7667	-0.0320	$[-0.0378, -0.0262]$
MMTIT	35B-A3B	0.7845	0.7807	$+0.0038$	$[-0.0005, +0.0079]$
FLORES-rendered	2B	0.5104	0.5221	-0.0117	$[-0.0223, -0.0012]$
FLORES-rendered	9B	0.6780	0.6714	$+0.0066$	$[-0.0018, +0.0147]$
FLORES-rendered	35B-A3B	0.7050	0.6926	$+0.0124$	$[+0.0036, +0.0217]$
Vistra (adapted)	2B	0.5505	0.6459	-0.0955	$[-0.1040, -0.0867]$
Vistra (adapted)	9B	0.6953	0.7143	-0.0190	$[-0.0265, -0.0122]$
Vistra (adapted)	35B-A3B	0.7168	0.7031	$+0.0137$	$[+0.0083, +0.0189]$

Table 6: Image-text translation COMET-22. Qwen3.5 columns use post-trained models; the Index 35B-A3B checkpoint is a preview. Δ is Index minus Qwen3.5. Paired bootstrap intervals cluster by image for MMTIT and Vistra and by source language for synthetic FLORES-rendered images. Vistra uses an adapted whole-image protocol; neither supplementary probe reports an official source-benchmark score.

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Despite using no images during mid- or post-training, Index-Translate achieves image-text translation performance comparable to corresponding post-trained Qwen3.5 models at all three scales. Larger models show stronger cross-modal generalization.

4.4 General capabilities

We evaluate general knowledge and multilingual question answering with C-Eval [21], GPQA-Diamond [22], INCLUDE [23], and MMMLU [24]. All models use the same evaluated subsets: 12,342 labeled C-Eval test questions across 52 subjects, 198 GPQA-Diamond questions, 22,639 INCLUDE questions in 44 languages, and 4,200 MMMLU questions sampled as 300 per language across 14 languages (seed 42). Scores are answer accuracy, with sample-level averaging for the multilingual sets.

Table 7 reports the results. Index-Translate-9B exceeds Hy-MT2-7B on all four benchmarks, while remaining below Qwen3.5-9B.

Model	C-Eval	GPQA-Diamond	INCLUDE	MMMLU
Index-Translate-35B-A3B (preview)	77.6	49.1	68.8	71.9
Index-Translate-9B	69.6	36.3	63.1	66.6
Index-Translate-2B	42.6	27.4	39.1	42.7
Hy-MT2-1.8B	46.0	25.7	37.6	40.1
Hy-MT2-7B	57.4	30.5	50.3	52.3
Hy-MT2-30B-A3B	78.0	48.9	67.4	70.3
North-Small-Translate	68.6	37.9	63.4	67.0
Qwen3.5-2B	62.8	44.0	40.5	47.9
Qwen3.5-9B	86.6	75.8	70.2	75.9
Qwen3.5-35B-A3B	89.8	**78.8**	76.7	83.0
DeepSeek-V4.1-Flash	**90.7**	75.6	82.0	83.6
GPT-5.6-Sol	87.5	72.5	84.5	**85.8**
Gemini 3.5 Flash Lite	88.2	74.0	**84.7**	85.0

Table 7: General-capability accuracy (%) on matched evaluation subsets. Index-Translate uses the same model versions as the text-translation results. INCLUDE and MMMLU are averaged over examples; MMMLU samples 300 questions per language. Bold marks the highest score per column.

4.5 Speech translation

Speech-to-text translation. The in-house video translation set contains 140 windows of 50–60 seconds, with 20 per direction: Chinese to English, Japanese, Korean, Spanish, Portuguese, and Arabic, plus English to Chinese. MT judge averages Gemini-3.1-Pro's per-line translation quality ratings on a 0/0.5/1 scale. ASR error is the median error rate between the model's transcription of the source audio and the reference transcript. Timestamp MAE is the mean absolute difference, in seconds, between predicted and reference sentence start times.

Index-Echo-2B and 9B achieve the two lowest ASR and timestamp errors (Table 8). The 9B translation score is 0.857, between Qwen3.8-Omni-Flash and Gemini-3.1-Pro with thinking. Qwen3.8-LiveTranslate is evaluated in streaming mode.

Speech-to-speech translation. We compare E2E dubbing with Index-Echo-S2TT + CosyVoice3 on in-house video dubbing data across six directions and two model sizes (Table 9). Each pair shares the same frozen Index-Echo-S2TT model and text translation, judged by Gemini-3.1-Pro. Chinese-source tests contain 100 examples for English and 200 each for Spanish and Japanese. English-source tests contain 200 examples each for Chinese, Spanish, and Japanese, with 146–198 retained after filtering. Content error is reported as mean / median, using WER for English and Spanish, normalized CER for Chinese, and katakana CER for Japanese. Speaker similarity is cosine similarity against the source speaker.

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Model	MT judge ↑	ASR error (median) ↓	Start-time MAE (s) ↓
Qwen3.8-Omni-Flash	**0.887**	0.112	1.004
Index-Echo-9B	0.857	0.102	0.488
Gemini-3.1-Pro (thinking)	0.849	0.169	1.824
Qwen3.8-LiveTranslate	0.833	0.186	–
Index-Echo-2B	0.825	**0.088**	**0.395**
Gemini-2.5-Flash (thinking)	0.817	0.201	2.402
Gemini-2.5-Flash	0.757	0.479	1.746
Qwen3-Omni	0.700	0.139	1.145
FireRed Audio	0.448	0.128	0.765
SeamlessM4T-v2	0.060	1.000	–

Table 8: S2TT comparison on the in-house video translation test set. The (thinking) label identifies baselines evaluated with thinking enabled. Dashes denote unavailable timestamp outputs. SeamlessM4T-v2 lacks line-level timestamp instructions and receives unchunked approximately 57-second inputs, beyond its approximately 30-second effective range.

E2E lowers mean content error in 8 of 12 size–direction pairs; Chinese-to-English favors the pipeline. Speaker similarity differences are small, with a 0.021 E2E gain for Chinese-to-English and absolute differences within 0.01 elsewhere. In an earlier Chinese-source evaluation, Index-Echo-2B also exceeds SeamlessM4T-v2 in MT judge score: 0.905 versus 0.370 for English, 0.823 versus 0.343 for Spanish, and 0.838 versus 0.310 for Japanese.

			Content error ↓ (mean / median)		Spk. ↑	
Direction / metric	Size	Text judge ↑	Pipeline	E2E	Pipeline	E2E
zh→en / WER	2B	0.830	**0.0450** / 0.000	0.0709 / 0.029	0.722	0.743
zh→en / WER	9B	0.840	**0.0477** / 0.000	0.0615 / 0.000	0.718	0.739
zh→es / WER	2B	0.728	0.0811 / 0.070	**0.0704** / 0.051	0.747	0.743
zh→es / WER	9B	0.793	0.0916 / 0.071	**0.0773** / 0.057	0.746	0.746
zh→ja / Kata CER	2B	0.760	0.0575 / 0.034	**0.0493** / 0.027	0.777	0.780
zh→ja / Kata CER	9B	0.815	0.0552 / 0.029	**0.0436** / 0.021	0.778	0.782
en→zh / CER	2B	0.825	0.0614 / 0.000	**0.0372** / 0.000	0.630	0.623
en→zh / CER	9B	0.800	**0.0463** / 0.000	0.0485 / 0.000	0.632	0.623
en→es / WER	2B	0.805	0.0867 / 0.070	**0.0702** / 0.000	0.732	0.735
en→es / WER	9B	0.820	0.0819 / 0.000	**0.0660** / 0.000	0.736	0.729
en→ja / Kata CER	2B	0.790	**0.0342** / 0.000	0.0345 / 0.000	0.707	0.698
en→ja / Kata CER	9B	0.803	0.0347 / 0.000	**0.0337** / 0.000	0.705	0.710

Table 9: Matched S2ST comparison on in-house video dubbing test data at 2B and 9B. Pipeline denotes Index-Echo-S2TT + CosyVoice3. Both systems share the same frozen Index-Echo-S2TT model and text translation. Bold marks the lower mean content error within each row. Spk. is source-speaker cosine similarity.

4.6 Syllable-controlled translation

SandGlass contains 300 subtitle sentences, with 60 each from animation, film and television, travel, gaming, and knowledge, and at most ten per video. Four target languages and three target syllable counts yield 3,600 cases per model. Central counts multiply subtitle duration by speaking rates of 6.19/7.84/7.82/8.4 syllables per second for English/Japanese/Spanish/Arabic; short and long targets scale these by 0.75 and 1.25. Local models use greedy decoding with a 512-token output limit; API baselines use their recorded settings. Gemini-2.5-Flash judges quality without a reference on a 0/0.5/1 scale, with the length constraint in the prompt. We report relative syllable deviation, rates within one syllable or 10% of the target, and output-versus-target regression slope across 1,200 three-budget groups (ideal: 1).

Index-Homura-9B and 2B achieve within-10% rates of 81.92% and 63.08%, exceeding the external baselines while reducing mean deviation (Table 10). DeepSeek-V4-Flash and GPT-5.6-Sol score higher on translation quality. Table 11 gives the language breakdown; Section 5.3 examines the quality–control trade-off. Appendix B.3 illustrates responses to different target counts.

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Model	Quality ↑	Mean $\lvert d\rvert$ ↓	Within ± 1 ↑	Within 10% ↑	Slope ≈ 1
Index-Homura-9B (RL)	0.7863	**0.0693**	**74.42%**	**81.92%**	**0.968**
Index-Homura-9B (SFT)	0.8581	0.1752	41.39%	45.75%	0.680
Index-Homura-2B (RL)	0.7615	0.0962	54.39%	63.08%	0.947
Index-Homura-2B (SFT)	0.7965	0.1594	37.89%	43.06%	0.672
GPT-5.6-Sol (low)	0.8715	0.3223	42.53%	47.64%	0.891
DeepSeek-V4-Flash (no thinking)	**0.8742**	0.3368	16.58%	16.36%	0.153
Hy-MT2-7B	0.8560	0.3174	18.14%	18.86%	0.195
Hy-MT2-30B-A3B	0.7554	0.2743	20.97%	22.39%	0.340
Hy-MT2-1.8B	0.5904	0.7080	14.58%	16.72%	0.222
Hunyuan-MT-7B	0.7188	1.0579	12.31%	15.11%	0.031
Qwen3.5-9B	0.7183	0.3247	15.61%	16.39%	0.430
Qwen3.5-35B-A3B	0.7863	0.3460	13.31%	13.25%	0.249
Qwen3.5-2B	0.4917	0.5151	12.50%	12.25%	0.036

Table 10: Expanded SandGlass evaluation on 3,600 cases per model. d is relative syllable-count deviation. Bold marks the best value among the displayed models (slope closest to 1). GPT-5.6-Sol uses the recorded gateway configuration, which may inject additional system prompts.

Target language	Cases	Quality ↑	Within 10% ↑
English	900	0.7950	78.67%
Japanese	900	0.8300	86.67%
Arabic	900	0.7661	79.89%
Spanish	900	0.7539	82.44%

Table 11: Index-Homura-9B by target language on SandGlass. Each language includes 300 sentences at three length targets.

4.7 Native long-document translation

Book data and benchmark. NativeLongBench evaluates native long-document translation on GuoFeng [16], BWB [25], and General Books. We select and align bilingual book material from these corpora. GuoFeng and BWB contain web fiction originally written in Chinese, with English translations. General Books adds English originals across literature, history, social science, science and technology, paired with model-generated Chinese translations. We preserve chapter order, align bilingual passages, and filter incomplete or mismatched pairs. Table 12 summarizes the books used in this study. Japanese originals paired with model-generated Chinese translations provide additional training supervision.

All three corpora are divided by work into training, validation, and test sets. GuoFeng test works are selected from its public training corpus because the official test set is not public. We also exclude identified copies of test works in other corpora from training. The benchmark draws continuous passages from held-out books, evaluating Chinese to English on GuoFeng/BWB and English to Chinese on General Books. This GuoFeng evaluation is distinct from the chapter-level test in Table 3.

Corpus	Original	Books	Train	Valid.	Test	Mean (K)	P25–P75 (K)
GuoFeng	Chinese	140	130	4	6	171.6	56.7–87.7
BWB	Chinese	337	327	4	6	777.2	277.0–1205.4
General Books	English	306	292	3	11	103.2	72.3–127.5

Table 12: Books used in this study. Lengths describe the available original-language text per training book (K = 1,024 tokens); P25–P75 denotes the middle 50% of the length distribution.

We plan to make the shareable portions of NativeLongBench publicly available.

Evaluation. The main comparison evaluates all models in the native setting: each receives the complete test document and is asked to translate it in a single generation. All test length groups use Chinese-side token counts: Chinese source text for GuoFeng/BWB and Chinese reference translations for General Books. The five groups span 4K–64K; the General Books 64K group corresponds to approximately 68K–82K English input tokens.

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These fixed evaluation sets have also been used during model development. Index-NativeLong uses greedy decoding with a 256K-token context window.

Gemini 3.8 Flash is excluded from the main comparison because its 65,536-token output limit [31] falls below the 128K output budget reserved for the longest test group. North's long-input results use an experimental extension beyond its supported 16K input/output range. Some GLM requests are refused; the affected results are marked in Table 13.

Scoring. We use SEGALE [26] to align the generated translation with the source before applying COMET [27]. The output is segmented into sentences, and semantic alignment matches contiguous source and output blocks. Each matched output block is paired with its source text and the corresponding reference translation, allowing sentence-level COMET to evaluate long documents despite differing sentence boundaries. Unmatched blocks receive zero. We average block scores within each document, then document scores within each length group, and finally weight the five groups equally per corpus.

Results. Index-NativeLong supports the full tested length range and achieves strong quality under native generation. The 9B model has the highest five-group means in Table 13 on GuoFeng, BWB, and General Books: 0.7891, 0.7683, and 0.8848. Several models score similarly on short passages, but their quality diverges as length increases (Figure 4). On GuoFeng, Qwen3.8 Flash falls from 0.7974 at 4K to 0.4939 at 64K, while Index-NativeLong-9B changes from 0.7836 to 0.7382. The 2B model also exceeds the external models at 32K and 64K on BWB. These results show that compact models adapted through document training can outperform existing translation and frontier models on native long-document translation.

The outputs reveal different failure modes. Hy-MT2 sometimes stops after a short fragment; North can degenerate into repetition without stopping on long inputs; Flash can leave sentences unfinished or retain source-language text. Some 2B outputs repeat passages, possibly reflecting the limits of model capacity during sustained generation.

Model	4K	8K	16K	32K	64K	Mean
GuoFeng (zh→en)						
Qwen3.8 Flash	**0.7974**	0.7956	0.7572	0.6826	0.4939	0.7053
GLM-5.3-Flash	0.7931	0.7957	0.7921	0.6018	0.5444	0.7054
North-Small-Translate-1.0	0.7952	0.7914	0.7309	0.5272	0.2353	0.6160
Hy-MT2-7B	0.7334	0.2872	0.0197	0.0006	0.0001	0.2082
Hy-MT2-30B-A3B	0.7535	0.4566	0.1615	0.0142	0.0664	0.2904
Index-NativeLong-2B	0.7704	0.7796	0.6508	0.6036	0.5460	0.6701
Index-NativeLong-9B	0.7836	**0.8074**	**0.8080**	**0.8084**	**0.7382**	**0.7891**
BWB (zh→en)						
Qwen3.8 Flash	0.7683	0.7676	0.7387	0.6262	0.5106	0.6823
GLM-5.3-Flash	0.7632	0.7622	0.7107	0.6344	0.4748	0.6691
North-Small-Translate-1.0	0.7717	**0.7704**	0.7715	0.6077	0.2196	0.6282
Hy-MT2-7B	0.7167	0.2387	0.0334	0.0024	0.0001	0.1983
Hy-MT2-30B-A3B	0.7301	0.4408	0.1432	0.0746	0.1103	0.2998
Index-NativeLong-2B	0.7688	0.7669	0.7201	0.7100	0.6843	0.7300
Index-NativeLong-9B	**0.7733**	0.7695	**0.7790**	**0.7756**	**0.7440**	**0.7683**
General Books (en→zh)						
Qwen3.8 Flash	0.8945	0.8910	0.8837	0.8582	0.5756	0.8206
GLM-5.3-Flash*	**0.9045**	0.8607	**0.8974**	0.8478	**0.8871**	0.8795
North-Small-Translate-1.0	0.8970	**0.8952**	0.8627	0.7873	0.7330	0.8350
Hy-MT2-7B	0.8246	0.3157	0.0335	0.0004	0.0057	0.2360
Hy-MT2-30B-A3B	0.8607	0.4395	0.1467	0.0097	0.1962	0.3306
Index-NativeLong-2B	0.8908	0.8736	0.8735	0.8435	0.8439	0.8651
Index-NativeLong-9B	0.9016	0.8771	0.8958	**0.8980**	0.8514	**0.8848**

Table 13: Native long-document translation with document-SEGALE/COMET (`max_size=8`). Mean equally weights the five Chinese-side length groups. Bold marks the highest displayed score in each column. *Some GLM samples were refused; its General Books scores average the remaining samples.

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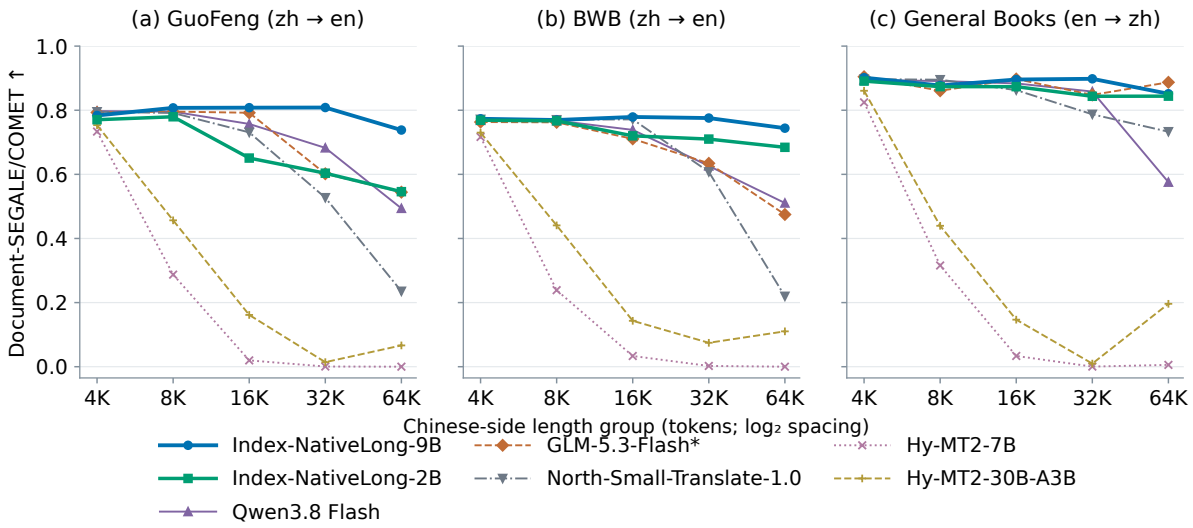


Figure 4: Native long-document translation across Chinese-side length groups, reproducing the point estimates in Table 13. Higher document-SEGALE/COMET is better; unmatched blocks receive zero, so the score reflects coverage as well as translation quality. *GLM's General Books scores average non-refused samples. North uses the experimental context extension described in Section 4.7.

4.8 Index-Translate-Omni

Text translation. Table 14 reports the existing evaluations of the text backbones used by the Omni releases. The 2B and 9B results come from the corresponding Index-Translate backbones in Table 4. The 35B-A3B release uses a later text backbone, with the published 35B-A3B preview shown for comparison.

Speech. Table 15 reports ASR character error rate (CER) and speech translation BLEU, chrF2, and target CER. Each evaluated subset has 30 examples per task, except for the 2B English-to-Chinese subset, which has 29. Cross-size comparisons should account for differences between the evaluated subsets. For 35B-A3B, the output limit is 256 tokens and English ASR uses a repetition penalty of 1.1. The other three tasks use the original decoding settings. One English-to-Chinese output reached the limit without yielding a target translation and remains in the score. On the same 30 inputs per task, the released 35B-A3B configuration lowers Chinese CER by 4.80 percentage points relative to its native audio baseline, raises English CER by 0.30 points, and changes English-to-Chinese and Chinese-to-English BLEU by $+2.54$ and -1.42, respectively. These differences reflect the evaluated weights and decoding settings together.

Image translation. Table 16 compares each Omni model with the same-size Qwen3.5 model on matching inputs. The scores come from the evaluated image configurations. The 35B-A3B entry reuses the prior image evaluation for the same language and vision weights, with 601 scored inputs and 73 unresolved cases outside the paired comparison. The speech and image results describe these evaluated subsets.

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Model	FLORES COMET-22	WMT24++ COMET-22	WMT26 Judge	MEME
Omni 2B	0.8655	0.8489	60.26	0.6443
Omni 9B	0.8789	0.8601	75.35	0.7387
Index-Translate 35B-A3B (preview)	0.8794	0.8586	76.76	0.7405
Omni 35B-A3B	0.8795	0.8593	78.01	0.7405

Model	instTrans Quality / IFscore	instTrans_minor Quality / IFscore / Off-target	IFMTBench XCOMET-XXL / IFscore
Omni 2B	0.5391 / 0.7569	0.3050 / 0.6586 / 3.97%	0.7712 / 0.7584
Omni 9B	0.6771 / 0.8209	0.5222 / 0.7725 / 3.47%	0.7957 / 0.8760
Index-Translate 35B-A3B (preview)	0.6901 / 0.8336	0.5151 / 0.7715 / 4.05%	0.7926 / 0.8991
Omni 35B-A3B	0.6867 / 0.8371	0.5163 / 0.7843 / 3.62%	0.7934 / 0.9070

Table 14: Existing text translation and instruction-following results for the Omni text backbones and the 35B-A3B preview reference. Lower off-target rates and higher values for the other metrics are better.

Model	Chinese ASR CER (%)	English ASR CER (%)	English → Chinese BLEU / chrF2 / CER (%)	Chinese → English BLEU / chrF2 / CER (%)
Omni 2B	13.8113	34.2560	36.982 / 30.357 / 43.3452	30.747 / 51.152 / 48.0670
Omni 9B	17.8917	14.4097	43.507 / 36.312 / 41.3767	34.656 / 55.376 / 46.3720
Omni 35B-A3B	13.0053	15.4950	40.074 / 32.862 / 44.8097	32.411 / 53.452 / 46.1700

Table 15: Omni speech recognition and speech-to-text translation on the evaluated subsets. Lower CER and higher BLEU and chrF2 are better.

Model size	MMTIT	FLORES-rendered	Vistra
2B	$52.164 \rightarrow 65.809$ (249)	$50.187 \rightarrow 53.612$ (120)	$55.326 \rightarrow 65.596$ (305)
9B	$73.814 \rightarrow 77.604$ (248)	$67.804 \rightarrow 68.438$ (119)	$68.963 \rightarrow 71.815$ (305)
35B-A3B	$75.602 \rightarrow 77.011$ (191)	$70.005 \rightarrow 69.891$ (118)	$71.067 \rightarrow 71.482$ (292)

Table 16: Paired image translation scores (COMET-22 $\times 100$, higher is better). Each cell shows the Qwen3.5 score, the corresponding Omni score, and the number of paired inputs.

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5 Analysis

5.1 Mid-training design and language coverage

The pilot studies explore data balance and parallel organization using greedy decoding. We use G for general text, M for independently sampled monolingual text, P for ordinary parallel data, and V for pivot-organized data.

For the constant-stage mixture sweep and the stage-progression study, translation scores take the equal-weight average of four dataset/source-language COMET means: FLORES-200 and WMT24++, each with English and Chinese sources. The constant-stage sweep covers 20 FLORES-200 targets and 18 WMT24++ targets. The decay studies use the evaluation scopes specified below and in Appendix A.2; their translation scores are compared within each study. The 11-task NLU score is the macro-average over MMMLU, Belebele, M3Exam, XCOPA, XCODAH, XCSQA, XStoryCloze, XWinogrande, M_ARC, MGSM, and INCLUDE, covering multilingual knowledge, reading comprehension, commonsense, and reasoning.

Mixture weights. The balanced $G{:}M{:}P = 1{:}1{:}1$ constant mixture gives the highest translation score among eight candidates and ties for the highest NLU score. Table 17 shows four settings, and Appendix A.2 supplies the other four from the same sweep. This pilot supports retaining both monolingual and parallel data. The appendix also reports a separate 4K decay-mixture study, whose translation scores are closely grouped.

Parallel organization. The Qwen3-8B parallel/pivot pilot uses a 16K context and compares $G{:}M{:}V = 1{:}2{:}4$ with $G{:}M{:}P{:}V = 1{:}2{:}2{:}4$ at step 10,000 and the same 25.17B-token decay budget (Table 17). Both mixtures include general and monolingual text. The first mixture contains no ordinary parallel data P; the second has $P{:}V = 1{:}2$ within the full four-stream mixture. On the 20-target-language translation evaluation, the mixture without P scores 0.7991, compared with 0.7971 for the mixture containing P. The observed gap of 0.0020 compares the two complete mixtures: including P also changes the normalized sampling shares of G, M, and V. The reported NLU scores are similar. This comparison and the 4K sampling-weight study in Appendix A.2 use different translation evaluation scopes.

Setting	Translation	11-task NLU
Constant-stage sampling weights ($G{:}M{:}P$), 7k steps		
$1{:}1{:}0$	0.7750	0.7655
$1{:}0{:}1$	0.7897	0.7637
$1{:}1{:}1$	**0.7914**	**0.7725**
$1{:}1{:}4$	0.7907	0.7644
Decay-stage sampling weights, 16K context, 25.17B tokens		
$G{:}M{:}V = 1{:}2{:}4$	**0.7991**	0.7526
$G{:}M{:}P{:}V = 1{:}2{:}2{:}4$	0.7971	**0.7554**

Table 17: Selected mid-training pilots. The constant panel belongs to the eight-setting sweep completed in Appendix A.2. The parallel/pivot panel uses Qwen3-8B, a 16K context, and a 20-target-language translation evaluation; both runs reach step 10,000 at the same decay budget. Both settings include G and M; the first contains no ordinary parallel data P. Its translation scores use a different evaluation scope from the appendix decay panel. Bold marks panel maxima.

Stage progression. Translation improves through both the constant and decay stages at 2B and 9B in the 20+-language development study (Table 18). General-task NLU drops during the constant stage and partially recovers during decay, remaining below the base-model score at both sizes.

Language expansion. Expanding the 2B decay data from 20+ to 100+ languages improves the non-core mean from 0.5605 to 0.6570, with an approximately unchanged core-language mean (Table 19). All 61 non-core target languages improve, and their off-target rate falls from 1.94% to 0.89%.

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	Translation			11-task NLU		
Model	Base	7k	10k	Base	7k	10k
Qwen3.5-2B	0.7015	0.7707	**0.7898**	**0.5920**	0.5303	0.5431
Qwen3.5-9B	0.7726	0.7924	**0.8007**	**0.7774**	0.7682	0.7712

Table 18: Training-stage progression in the 20+-language development study, using the metrics defined in Section 5.1.

Backbone / mid-training	Core 20	Non-core 61	All 81
2B base	0.7319	0.5172	0.5702
2B, 20+ languages	0.8231	0.5605	0.6253
2B, 100+ languages	0.8219	**0.6570**	**0.6977**
9B base	0.8059	0.6952	0.7225
9B, 20+ languages	0.8339	0.7141	0.7437

Table 19: Language expansion at the mid-training stage on 81 target languages. Corrected COMET averages English- and Chinese-source directions per target, assigning zero to off-target outputs.

Broad language coverage during SFT. In a 22-language-only SFT control, 60-language FLORES-200 COMET falls from 0.7441 to 0.7062 between steps 200 and 400, and target-language agreement falls from 59.22% to 40.02%, while core-language scores remain nearly unchanged. This failure motivates retaining low-resource directions during SFT and evaluating language coverage beyond the core set.

5.2 Specialist training and integration

SFT-to-RL gains at 9B. Table 20 compares each specialist before and after RL. For the Instruction expert, instTrans quality improves by 0.0386 and IFscore by 0.0934, with gains on IFMTBench and low-resource instructions. For the General expert, low-resource FLORES COMET-22 improves by 0.0236 and off-target rate falls from 3.4% to 1.2%. FLORES XCOMET-XXL and both WMT24++ metrics improve, while FLORES COMET-22 decreases by 0.0017. These results show stronger instruction adherence and low-resource translation after RL, with metric-dependent changes on core-language translation.

Metric	SFT	RL	Δ (RL $-$ SFT)
Instruction specialist, 9B			
instTrans quality	0.6320	0.6706	+0.0386
instTrans IFscore	0.7670	0.8604	+0.0934
IFMTBench XCOMET-XXL	0.7804	0.7941	+0.0137
IFMTBench IFscore	0.8662	0.8977	+0.0315
Low-resource instTrans quality	0.4817	0.5056	+0.0239
Low-resource instTrans IFscore	0.7502	0.7894	+0.0392
General specialist, 9B			
FLORES COMET-22	0.8850	0.8833	-0.0017
FLORES XCOMET-XXL	0.9092	0.9140	+0.0048
WMT24++ COMET-22	0.8574	0.8592	+0.0018
WMT24++ XCOMET-XXL	0.8840	0.8942	+0.0102
Low-resource FLORES COMET-22	0.8129	0.8365	+0.0236
Low-resource FLORES XCOMET-XXL	0.6964	0.7321	+0.0357
Low-resource off-target rate	3.4%	1.2%	-2.2 pp

Table 20: SFT-to-RL changes for the 9B Instruction and General specialists before model integration. Each RL model is compared with its own SFT baseline. Instruction evaluation uses GPT-5.6-Sol. Score differences are absolute; off-target changes are percentage points (pp), where lower is better.

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Parameter interpolation. These ablations use an earlier instruction-expert configuration at the same interpolation weights; the main results use the updated instruction specialist. Instruction scores use GPT-5.6-Sol. The reference blend improves FLORES and WMT XCOMET-XXL at both sizes, while IFMTBench IFscore declines relative to the general parent; MEME improves at 2B and decreases slightly at 9B (Table 21). A separate 2B, 20-language weight scan supports retaining a dominant general branch: reducing its weight from 1.0 to 0.5 raises WMT XCOMET-XXL from 0.8222 to 0.8522 but lowers IFscore from 0.7395 to 0.2828.

Size	Model	FLORES X	WMT X	IFMT IF	MEME
2B	General	0.8408	0.8336	**0.7554**	0.6034
2B	$0.8G + 0.1\,\text{Instruction} + 0.1M$	**0.8566**	**0.8419**	0.7241	**0.6429**
9B	General	0.8972	0.8785	**0.8757**	**0.7383**
9B	$0.8G + 0.1\,\text{Instruction} + 0.1M$	**0.9025**	**0.8819**	0.8524	0.7331

Table 21: Expert integration ablations in the 100+-language setting, using the reference instruction-expert configuration. G denotes General and M denotes Meme. X denotes XCOMET-XXL; IF denotes IFMTBench IFscore. Bold marks the higher value within each model size.

Syllable control also transfers poorly in a four-way 2B merge: a 0.1-weight video-translation expert yields a SandGlass hit rate of 0.1544, compared with 0.5439 for the specialist.

Targeted MOPD enhancement. Larger teachers yield the best instTrans and IFMTBench instruction scores among the three MOPD variants, while all variants reduce low-resource off-target rates from 4.2% to 2.2% (Table 22). These gains come with lower translation quality and MEME scores, supporting targeted use for instruction adherence and language consistency.

	instTrans		IFMTBench	Low-resource pairs		MEME
Model / mixture	Quality ↑	IF ↑	IF ↑	COMET ↑	Off-target ↓	Score ↑
Reference merge	**0.5979**	0.8359	0.7241	0.7331	4.2%	**0.6429**
MOPD (2B, 1:1:1)	0.4848	0.8498	0.7448	0.7511	**2.2%**	0.6150
MOPD (2B, 1:2:1)	0.5079	0.8503	0.7470	**0.7564**	**2.2%**	0.6068
MOPD (9B→2B, 1:1:1)	0.5572	**0.8669**	**0.7504**	0.7516	**2.2%**	0.6275

Table 22: Targeted MOPD for the reference parameter-merge configuration. All evaluated models have 2B parameters. Mixtures are General:Instruction:Meme; 9B→2B denotes the larger-teacher configuration. IF is IFscore. Low-resource pairs are evaluated on FLORES with COMET-22 and off-target rate. Bold marks the best value per column.

We find that MOPD performs poorly at following length-related constraints. We hypothesize that this limitation may be related to its token-level supervision, and leave further investigation and improved constraint handling to future work.

5.3 Instruction and length constraints

Constraint composition. Index-Translate-9B scores above 2B in all twelve IFMTBench categories, with the largest gaps on constraints combining context and terminology (Table 23). Both models score lower on the combined-constraint subset than on single constraints; all five combined categories include terminology.

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Constraint	Items	2B	9B
Single constraints			
Code tags	161	0.9938	1.0000
Inline code	224	0.8929	1.0000
Layout	546	0.9579	0.9689
Structured data	392	0.9413	0.9974
Style	769	0.8252	0.8908
Terminology	1,364	0.7109	0.8364
Context	770	0.6522	0.8151
Weighted mean	4,226	0.7947	0.8894
Combined constraints			
Structured data + terminology	567	0.8713	0.9365
Style + terminology	525	0.8179	0.9177
Context + style + terminology	619	0.6532	0.8635
Structured data + terminology + style	637	0.6063	0.7717
Context + terminology	490	0.5808	0.7967
Weighted mean	2,838	0.7042	0.8560

Table 23: IFMTBench instruction-following scores for the final Index-Translate models, using GPT-5.6-Sol. Item counts describe the cleaned evaluation subset (7,064 examples); the overall instruction-following score averages 7,063 valid judgments. Section 4.1 explains preprocessing and scoring coverage.

Context and syllable control. On low-resource instTrans, translation quality is 0.3050 for 2B and 0.5222 for 9B, while IFscore is 0.6586 and 0.7725. On FLORES low-resource pairs, XCOMET-XXL is 0.4817 for 2B and 0.6805 for 9B, with off-target rates of 4.2% and 4.0%. Index-Translate-35B-A3B (preview) obtains 0.5151 quality and 0.7715 IFscore on low-resource instTrans, with a 4.05% off-target rate. On FLORES low-resource pairs, it scores 0.8168 COMET-22 and 0.7164 XCOMET-XXL, with a 2.4% off-target rate. The 9B model improves cross-sentence terminology and coreference resolution in low-resource directions (Table 24). In core directions, subtitle syllable alignment passes in 46.6% of 2B cases and 64.1% of 9B cases, while the other hard-constraint pass rates exceed 93%. This source-relative subtitle constraint complements the explicit target-count control studied with Index-Homura.

		Core directions		Low-resource directions	
Constraint	Items (core / low)	2B	9B	2B	9B
Subtitle syllable alignment	601 / 765	46.6%	64.1%	72.8%	84.6%
Style consistency	633 / 988	0.7157	0.8525	0.5220	0.7364
Cross-sentence terminology	508 / 777	0.6486	0.7139	0.4027	0.6536
Coreference resolution	24 / 546	0.6667	0.6250	0.5250	0.7542
Context disambiguation	42 / 54	0.6026	0.7439	0.5566	0.7593

Table 24: Selected constraints in instTrans and its low-resource extension. Syllable alignment is a binary-check pass rate; the remaining rows are mean soft-constraint scores from GPT-5.6-Sol. Item counts are per constraint, and a sample can contain multiple constraints.

Quality–control trade-off. Increasing the 2B syllable-reward weight from 1 to 2 raises the SandGlass within-10% rate from 63.08% to 74.19%, while translation quality falls from 0.7615 to 0.6900. Appendix B provides representative instruction-translation cases.

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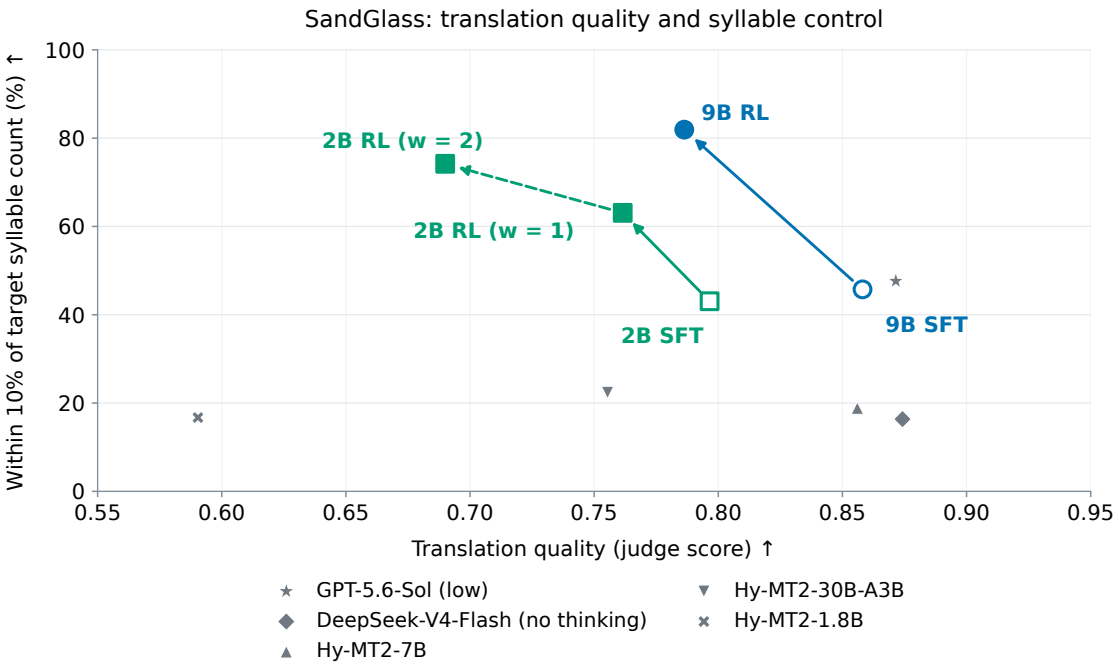


Figure 5: Quality and syllable-control trade-offs on SandGlass. Points reproduce Index-Homura and selected baselines from Table 10; the additional 2B point with syllable-reward weight $w = 2$ comes from the study in this section. Blue and green denote the 9B and 2B Homura models. Solid arrows connect SFT to RL, and the dashed arrow connects the two 2B reward weights. Both axes favor larger values; arrows show changes between evaluated configurations.

6 Conclusion

Index-Translate is a multilingual translation model family that combines multilingual mid-training and specialist integration for general translation and instruction following across 150 languages. Our experiments highlight the importance of broad language coverage and the trade-offs between translation quality and constraint adherence. Text-only mid- and post-training also support image-text translation through the visual pathway inherited from Qwen3.5 base models, with stronger cross-modal generalization at larger scales. Index-Echo, Index-Homura, and Index-NativeLong extend the shared foundation through dedicated training for speech, syllable control, and native long-document translation.

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A Supplementary Training Studies

A.1 Parallel and pivot organization

Let x_ℓ denote the same content expressed in language ℓ. Ordinary parallel data align a language pair directly. Pivot organization connects multiple translations through a shared English anchor, as illustrated in Table 25. Parallel degree counts aligned language variants, whereas $P{:}V$ specifies the sampling ratio between the two data streams.

Organization	Illustrative alignment	Parallel degree
Parallel (P)	$x_{\mathrm{zh}} \longleftrightarrow x_{\mathrm{ja}}$	2 languages
Pivot (V)	$x_{\mathrm{zh}} \longleftrightarrow x_{\mathrm{en}} \longleftrightarrow x_{\mathrm{ja}}$	3 languages

Table 25: Illustrative parallel and pivot groups. Arrows denote cross-language alignment; the pivot example shares one English anchor across the Chinese and Japanese variants.

A.2 Additional mid-training mixtures

The constant-stage rows in Table 26 complete the same eight-setting $G{:}M{:}P$ sweep at step 7,000 as the upper panel of Table 17. G, M, P, and V denote general, monolingual, parallel, and pivot data.

The decay rows are a separate 4K sampling-weight study at step 10,000, varying how pivot data are mixed with general and monolingual text. Their translation score takes the equal-weight average of six dataset/source-language COMET means: WMT24++, FLORES-200, and BOUQuET, each with English and Chinese sources and eight target languages. The $G{:}M{:}V = 1{:}2{:}4$ row uses the same sampling weights as the setting without ordinary parallel data in Table 17, but the latter uses a 16K context and reports translation over 20 target languages. Translation scores should therefore be compared within each study. Each panel compares configurations at the same training step; stage-to-stage trajectories are reported separately in Table 18.

Phase	Sampling weights	Translation	11-task NLU
Constant, 4K, 7k	$G{:}M{:}P = 1{:}1{:}2$	0.7903	0.7638
	$G{:}M{:}P = 1{:}2{:}1$	0.7898	0.7711
	$G{:}M{:}P = 2{:}1{:}1$	0.7873	0.7641
	$G{:}M{:}P = 2{:}2{:}1$	0.7879	0.7725
Decay, 4K, 10k	$G{:}M{:}V = 1{:}2{:}2$	0.8023	0.7763
	$G{:}M{:}V = 1{:}2{:}4$	0.8033	0.7717
	$G{:}V = 1{:}8$	0.8019	0.7702
	$G{:}V = 1{:}4$	0.8027	0.7710

Table 26: Additional pilot mixtures. Constant rows complete the sweep in Table 17. Decay rows form a separate 4K study using the eight-target-language, six-component translation mean defined in Appendix A.2. NLU is the 11-task macro-average defined in Section 5.1. Compare translation scores within each study.

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A.3 NativeLong training ablations

We study decay sequence length, position encoding, and length generalization in separate 2B experiments using GuoFeng-only document SFT (Table 27). Scores use the same 0–1 scale as the main evaluation; each study averages over its own evaluation passages.

Long-sequence decay. With matched short document SFT, using prepacked 128K rather than 4K sequences during mid-training decay improves mean quality by 0.0528 with RoPE [11] and 0.0495 with NoPE (no explicit positional encoding) [10]. The 128K route uses exponential learning-rate decay under WSD.

Position encoding and length generalization. After document SFT covering source lengths up to approximately 64K, the two 128K-decay bases have similar overall scores: approximately 0.7810 for RoPE and 0.7819 for NoPE. Their strengths vary by length, with NoPE ahead at 16K and RoPE ahead at 64K. When document SFT is limited to source texts of approximately 16K or less, NoPE scores higher over the 4K–32K evaluation range, with a 0.0370 advantage at 32K.

(a) Decay sequence length, followed by matched short document SFT

Position encoding	4K decay	128K decay	Gain
RoPE	0.6673	**0.7201**	+0.0528
NoPE	0.6739	**0.7234**	+0.0495

	(b) Document SFT up to 64K		*(c) Document SFT up to 16K*	
Test length	RoPE	NoPE	RoPE	NoPE
4K	0.7930	**0.7950**	0.7910	**0.7918**
8K	**0.8000**	0.7970	0.7839	**0.7974**
16K	0.6780	**0.7910**	0.7318	**0.7586**
32K	**0.8050**	0.7920	0.6264	**0.6634**
64K	**0.7670**	0.6780	–	–
Overall	≈ 0.7810	$\approx \mathbf{0.7819}$	0.7513	**0.7667**

Table 27: GuoFeng 2B ablations using document-SEGALE/COMET (`max_size`=8) on a 0–1 scale. Panels (b) and (c) start from 128K-decay bases. Each overall score averages the evaluation passages in its own study.

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B Supplementary Evaluation Details and Cases

B.1 Translation with task constraints

We present two examples from the instruction-following evaluations, showing the source, task requirements, and recorded Index-Translate-9B output.

Cross-sentence terminology. The Chinese-to-Russian subtitle task asks the model to use consistent translations for cracks, carbon fiber, and concrete. Table 28 shows three segments in which the translation preserves the technical vocabulary, with grammatical inflection across sentences.

Segment	Source and Index-Translate-9B translation
2	王平仲却采用了，更加昂贵的碳纤维材料 Ван Пинчжун решил использовать более дорогой материал — углеродное волокно.
5	碳纤维的好处呢，就是它抗裂缝 Преимущество углеродного волокна в том, что оно противостоит трещинам.
6	而且混凝土的刚性，老的混凝土以前不是裂缝嘛 К тому же, что касается жёсткости бетона — ведь в старом бетоне раньше были трещины, не так ли?

Table 28: Chinese-to-Russian subtitle excerpts from instTrans. Three segments from the same ten-segment input illustrate terminology consistency; source text and translations are reproduced verbatim.

Structured-data translation. The English-to-Chinese task asks for translation of the Category, Description, and Release_Status columns, preserving the CSV header, field boundaries, product IDs, and prices. Table 29 reproduces the complete input and output.

Source (English)

```
Product_ID,Category,Description,Release_Status,Price
"SYS-001","Hardware","High-performance motherboard, equipped with dual Gigabit LAN and PCIe 4.0
slots.","Available",199.99
"SYS-002","Software","Enterprise-level firewall solution, providing real-time threat detection and
automated response protocols.","In Beta",49.99
```

Index-Translate-9B (Chinese)

```
Product_ID,Category,Description,Release_Status,Price
"SYS-001","硬件","高性能主板，配备双千兆局域网和PCIe 4.0插槽。","已发布",199.99
"SYS-002","软件","企业级防火墙解决方案，提供实时威胁检测和自动化响应协议。","测试版",49.99
```

Table 29: English-to-Chinese structured translation from IFMTBench. The header, product IDs, prices, and CSV field boundaries are preserved while the three requested text columns are translated.

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B.2 Meme translation cases

Table 30 illustrates nonstandard spelling, a community-specific alias, and a laughter expression. Both Index-Translate sizes recover the intended meanings in these selected examples.

Model	Recorded translation
1. Playful proposal. Source: 那个，你好，结芬<EMOTE-placeholder-0>	
Intended meaning: A playful request to get married, rather than a personal name.	
GPT-5.6-Sol	Um, hello, Jie Fen <EMOTE-placeholder-0>
Hy-MT2-7B	Hey there, Jiefen<EMOTE-placeholder-0>
Index-Translate-2B	Um, hi there, let's get married<EMOTE-placeholder-0>
Index-Translate-9B	Um, hi, let's get married <EMOTE-placeholder-0>
2. Gaming-community alias. Source: 狒瘾犯了就去打	
Intended meaning: An urge to play Final Fantasy XIV (FFXIV).	
GPT-5.6-Sol	If you're itching for more JoJo, go play the game.
Hy-MT2-7B	If you get monkey addiction, go fight.
Index-Translate-2B	Go play when my FFXIV addiction kicks in.
Index-Translate-9B	When the FFXIV itch hits, just go play.
3. Laughter expression. Source: Fuyuan Flower红红火火恍恍惚惚	
Intended meaning: The repeated pinyin initial h conveys laughter, like repeated "ha".	
GPT-5.6-Sol	Fuyuan Flower—red-hot and thriving, hazy and wavy.
Hy-MT2-7B	Fuyuan Flower: booming yet confused.
Index-Translate-2B	Fuyuan Flower, hahaha, what a wild ride!
Index-Translate-9B	Fuyuan Flower: Hahahahaha, this is wild!

Table 30: Selected MEME examples from the final Index-Translate models. Both Index-Translate sizes receive a judge score of 1 on each example; the displayed baselines receive 0. Intended meanings summarize benchmark annotations. Translations and placeholders are reproduced from the recorded outputs.

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B.3 Syllable-controlled translation cases

Table 31 shows Index-Homura-9B translating the same source at three target syllable counts. The English example varies introductory wording and noun phrases; the Japanese example uses increasingly expanded expressions for a deeply ingrained traumatic memory.

Budget	Target	Actual	Recorded translation
Chinese to English. Source: 说到底聊天群的规则一句话就能总结			
Short	10	10	The group's rules can be summed up in one line.
Natural	14	14	Ultimately, the group's rules can be summed up in one line.
Long	18	18	Ultimately, the rules of the chat group can be summed up in one sentence.
Chinese to Japanese. Source: 先前被虐的阴影可谓是印刻到了骨子里头了			
Short	15	15	虐められたトラウマが根深い。
Natural	20	20	前のトラウマが骨に刻まれてるんだな
Long	25	25	前の屈辱が骨の髄まで刻み込まれていたな。

Table 31: Selected Index-Homura-9B examples from SandGlass. Target and actual syllable counts follow the benchmark's counting procedure. Each displayed output exactly meets its target and receives a Gemini-2.5-Flash quality score of 1.0. Translations are reproduced verbatim.

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B.4 Long-document translation cases

Comparison setup. The three documents in Section 3.3 are held out by work. We compare native Index-NativeLong-9B with three TranslateBooksWithLLMs configurations (version 1.5.13): the same 9B model with an automatic glossary, Qwen3.8-Flash with an automatic glossary, and Flash with a glossary generated by GPT-6-Sol from the full source. Automatic glossaries are generated by the translating model from source excerpts sampled by the framework. All glossaries use source text alone. Chunked translation receives neighboring source text and the preceding translation's tail.

Glacial history: preserving the author's concepts. This English-to-Chinese example comes from the 32K group of General Books. Around 10.4% into the source, the author defines an *ice age* as a whole cold episode that includes both *glacials* and *interglacials*. Table 32 presents the source and four translations of this passage. The chunked 9B run uses temperature 0.2 and repetition penalty 1.05.

Configuration	Source or translation excerpt
Source	In current usage, the term "ice age" properly refers to an entire cold episode, including its short warm periods. […] Large-scale advances and retreats of ice during an ice age are usually referred to as glacials and interglacials respectively.
Native 9B	按当前用法，"冰河时代"一词应指整个寒冷事件，包括其中的短暂温暖期。[…]冰河时代中冰层的大规模进退，通常分别称为冰期和间冰期。
Chunked 9B + automatic glossary	按照当前用法，"冰河时代"一词应指整个寒冷时期，包括其中短暂的温暖期。[…]冰河时代中冰层的大规模推进和退缩，通常分别称为冰期和间冰期。
Chunked Flash + automatic glossary	在当今的用法中，"冰期"一词恰当地指代整个寒冷时期，包括其间短暂的温暖阶段。[…]在一个冰期内，大规模冰川推进和消退通常分别被称为冰期和间冰期。
Chunked Flash + full-source glossary	在当前的用法中，"冰期"一词确切地指代整个寒冷时期，包括其间短暂的温暖阶段。[…]在冰期内，大规模冰川的推进和消退通常分别被称为冰期和间冰期。

Table 32: Glacial-history excerpts at the same source passage. Bracketed ellipses omit intervening text. Both 9B translations preserve the whole–part distinction; both Flash translations use 冰期 for the whole episode and its cold stages.

The glossary conditions differ. The 9B automatic glossary contains *Ice Age* → Ice Age, leaving the term in English; Flash's automatic glossary omits *ice age*, *glacial*, and *interglacial*. The full-source glossary supplies *ice age* → 冰期, *glacial* → 冰期（冰进期）, and *interglacial* → 间冰期. Thus, the two Flash outputs lose the distinction under different glossary conditions: a fixed term list, even one prepared from the full source, does not ensure that the translation preserves the author's conceptual hierarchy.

Entity consistency. Table 33 pairs source excerpts with recurring names in the four translations. The early mention identifies 王妃 as a mentor. Her name is missing from the 9B glossary, mapped to Princess Consort in Flash's automatic glossary, and correctly supplied as Wang Fei by the full-source glossary. All three chunked glossaries omit Zeitlin.

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Configuration	Early mention	Middle mention	Late mention
Character identity: 王妃. Chinese to English; positions 31.3%, 56.6%, and 84.8%.			
Source excerpt	王妃导师让我来找你过去一趟	又被王妃叫去了办公室	王妃看到圣枪骑士的时候
Full-document 9B	Mentor Wang Fei	Wang Fei	Wang Fei
Chunked 9B + automatic glossary	Instructor Wangfei	the Dean	The Queen Consort
Chunked Flash + automatic glossary	Princess Consort	the Princess Consort	Princess Consort
Chunked Flash + full-source glossary	Instructor Wang Fei	Wang Fei	Wang Fei
Name consistency: Zeitlin. English to Chinese; positions 32.9%, 38.7%, and 68.2%.			
Source excerpt	a perspective Zeitlin has termed	Zeitlin and Tolli-day's work	As Zeitlin has reminded us
Full-document 9B	泽特林	泽特林	泽特林
Chunked 9B + automatic glossary	泽特林	Zeitlin	泽特林
Chunked Flash + automatic glossary	齐特林（Zeitlin）	齐特林	蔡特林（Zeitlin）
Chunked Flash + full-source glossary	齐特林	齐特林	蔡特林（Zeitlin）

Table 33: Selected entity mentions under four long-document translation configurations. Positions are measured by source-character offset; excerpts omit surrounding text. Dean and Queen Consort change the character's identity, while Wangfei and Wang Fei differ only in spacing.

Translation workflow. Across the three documents, native 9B uses three model calls and 90,143 input tokens, compared with 160 calls and 225,125 input tokens for the same model using automatic glossaries and chunked translation. The difference reflects extra calls and repeated context in the chunked workflow.

B.5 Judge failure cases

Policy outputs sometimes append unsupported markers such as `[1]` and `/ End/`, without adding source-grounded meaning (Table 34). Reward optimization can amplify evaluator errors, and adversarial suffixes can inflate LLM-judge scores [28, 29]. These observations motivate the adversarial judge supervision in Section 2.2.

Pattern	Source meaning	Recorded output
Citation suffix	It probably will not be a big event.	No será un gran acontecimiento. `[1]`
Termination marker	Pulled him out of that rubbish.	Lo saqué de entre toda esa porquería. `/ End/`

Table 34: Representative unsupported suffixes in policy outputs. Source meanings are English glosses of the Chinese inputs; Spanish outputs preserve the recorded markers.

B.6 IFMTBench preprocessing and scoring

The original IFMTBench contains 7,344 examples: 4,506 single-constraint and 2,838 combined-constraint items [15]. Our evaluation uses a cleaned subset of 7,064 examples (4,226 single-constraint and 2,838 combined-constraint items), after removing 280 single-constraint examples with data-quality issues. These include corrupted reference text, where unassigned or private-use code points replace or interrupt ordinary words, and inconsistencies between language labels and the source or reference language. The reported instruction-following aggregate averages 7,063 valid judgments; one retained example did not yield a valid judgment.

For combined constraints, hard checks gate the mean graded constraint score. IFscore is the sample-level mean over valid judgments, and reference-based XCOMET-XXL measures translation quality.

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C Supported Languages and Identifiers

The supported-language inventory contains 150 entries, including English and Chinese: 22 core and 128 additional entries. Regional and script-specific varieties are counted separately as listed. Four pairs of alternate codes each identify one entry, yielding 154 identifiers in total. Tables 35 and 36 provide the complete inventory with these alternate codes grouped together. Benchmark subsets are specified separately in Section 4.1.

Table 35: Core languages: 22 entries, including English and Chinese. Alternate identifiers are grouped together.

Code(s)	Language / variety	Code(s)	Language / variety
en, eng_Latn	English	zh, zho_Hans	Chinese
de	German	fr	French
es	Spanish	ja	Japanese
ko	Korean	pt	Portuguese
ru	Russian	ar	Arabic
it	Italian	nl	Dutch
pl	Polish	ro	Romanian
sv	Swedish	tr	Turkish
hi	Hindi	vi	Vietnamese
th	Thai	id	Indonesian
ms	Malay	fil	Filipino

Table 36: Additional languages and varieties: 128 entries. Alternate identifiers are grouped together.

Code(s)	Language / variety	Code(s)	Language / variety
ukr_Cyrl	Ukrainian	fas_Arab, pes_Arab	Persian
ces_Latn	Czech	ell_Grek	Greek
dan_Latn	Danish	hun_Latn	Hungarian
fin_Latn	Finnish	nob_Latn	Norwegian Bokmål
slk_Latn	Slovak	bul_Cyrl	Bulgarian
bos_Latn	Bosnian	cat_Latn	Catalan
ben_Beng	Bengali	heb_Hebr	Hebrew
lit_Latn	Lithuanian	slv_Latn	Slovenian
ekk_Latn	Estonian	als_Latn	Albanian
lvs_Latn	Latvian	azj_Latn	Azerbaijani
hrv_Latn	Croatian	tam_Taml	Tamil
npi_Deva	Nepali	urd_Arab	Urdu (Arabic script)
mkd_Cyrl	Macedonian	srp_Cyrl	Serbian (Cyrillic)
mar_Deva	Marathi	kat_Geor	Georgian
mal_Mlym	Malayalam	kaz_Cyrl	Kazakh
isl_Latn	Icelandic	glg_Latn	Galician
kan_Knda	Kannada	ary_Arab	Moroccan Arabic
guj_Gujr	Gujarati	bel_Cyrl	Belarusian
afr_Latn	Afrikaans	hye_Armn	Armenian
khk_Cyrl	Mongolian	khm_Khmr	Khmer
eus_Latn	Basque	mya_Mymr	Burmese
lat_Latn	Latin	uzn_Cyrl	Uzbek (Cyrillic)
tel_Telu	Telugu	ory_Orya	Odia
nno_Latn	Norwegian Nynorsk	uzn_Latn	Uzbek (Latin)
swh_Latn	Swahili	sin_Sinh	Sinhala
kir_Cyrl	Kyrgyz	som_Latn	Somali
pan_Guru	Punjabi	arz_Arab	Egyptian Arabic
cym_Latn	Welsh	nrm_Latn	Norman
pbt_Arab	Pashto	gle_Latn	Irish
srp_Latn	Serbian (Latin)	hau_Latn	Hausa

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Table 36 continued

Code(s)	Language / variety	Code(s)	Language / variety
ckb_Arab	Central Kurdish	mlt_Latn	Maltese
yue_Hani, yue_Hant	Cantonese	tgk_Cyrl	Tajik
kmr_Latn	Northern Kurdish	bew_Latn	Betawi
amh_Ethi	Amharic	lao_Laoo	Lao
ltz_Latn	Luxembourgish	fry_Latn	Frisian
div_Thaa	Dhivehi	epo_Latn	Esperanto
kin_Latn	Kinyarwanda	ars_Arab	Najdi Arabic
fao_Latn	Faroese	plt_Latn	Malagasy
asm_Beng	Assamese	snd_Arab	Sindhi
xho_Latn	Xhosa	tuk_Latn	Turkmen
hat_Latn	Haitian Creole	gla_Latn	Scottish Gaelic
ceb_Latn	Cebuano	ydd_Hebr	Yiddish
jav_Latn	Javanese	bak_Cyrl	Bashkir
pap_Latn	Papiamento	mri_Latn	Māori
uig_Arab	Uyghur	tat_Cyrl	Tatar
bod_Tibt	Tibetan	mww_Latn	White Hmong
hyw_Armn	Western Armenian	yor_Latn	Yoruba
zul_Latn	Zulu	sdh_Arab	Southern Kurdish
smo_Latn	Samoan	cos_Latn	Corsican
hif_Latn	Fiji Hindi	sun_Latn	Sundanese
nya_Latn	Chichewa	ibo_Latn	Igbo
haw_Latn	Hawaiian	sot_Latn	Sesotho
lus_Latn	Mizo	sna_Latn	Shona
chv_Cyrl	Chuvash	azb_Arab	South Azerbaijani
roh_Latn	Romansh	run_Latn	Kirundi
hin_Latn	Hindi (Latin)	sah_Cyrl	Yakut
tir_Ethi	Tigrinya	ast_Latn	Asturian
oci_Latn	Occitan	cnh_Latn	Hakha Chin
sme_Latn	Northern Sami	nds_Latn	Low German
oss_Cyrl	Ossetian	urd_Latn	Urdu (Latin)
gsw_Latn	Swiss German	anp_Deva	Angika
apc_Arab	Levantine Arabic	gaz_Latn	Oromo
nap_Latn	Neapolitan	hsb_Latn	Upper Sorbian
hil_Latn	Hiligaynon	kal_Latn	Greenlandic

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